# GridRescue AI · Financial Model — summary

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**Three separate money stories, never mixed:** (a) the hospital's savings · (b) the ESCO's project cash flow · (c) GridRescue AI's software revenue. Engine results are modelled (±30 %, to be confirmed with the hospital's load data); software prices are proposed and require market validation; the hospital's energy-as-a-service fee is the hospital's own cost, not GridRescue AI revenue.

## A · Hospital case — National Orthopaedic Hospital, Enugu

The hospital's savings: diesel avoided minus service costs. Engine results, modelled, ±30 %.

Inputs **PUBLISHED**: diesel ₦3,277/L (NBS price watch, national average, not an Enugu pump price) · grid ₦160/kWh (EERC Band A) · about 6 h of grid a day (reported, Brief 12) · 2.68 kg $CO_2$/L (IPCC default).

| PROTECTED LOAD | TODAY | WITH SYSTEM |
|---|---|---|
| Diesel, L/yr **ENGINE** | **1,042,790** | **176,112** |
| Operating cost, ₦ M/month: modelled diesel → service fee, ₦0 upfront **ENGINE** | **299.2** | **69.1** |
| … a year, ₦ bn **FORMULA** | 3.59 | 0.83 |
| Year-one energy cost, ₦ M/yr: grid + modelled diesel → grid + service fee **ENGINE** | **3,764.5** | **1,588.9** |
| 5 years, ₦ bn — saved 10.9 **FORMULA** | 18.8 | 7.9 |
| 10 years, ₦ bn — saved 21.8 **FORMULA** | 37.6 | 15.9 |
| $CO_2$, t/yr, Scope 1 + Scope 2 **ENGINE** | **2,795** | **549** |
| Theatre-hours a year without grid, of 12,480 **ENGINE** | **4,992 (40 %)** | **0** |
| Generator switchovers reaching equipment, a year **ENGINE** | **≈1,460** | **0** |

**Diesel avoided:** 866,678 L a year (83 %). **$CO_2$ avoided, net:** 2,323 t Scope 1 − 77 t Scope 2 = **2,246 t a year**; about 22,460 t over 10 years **FORMULA**.

**Break-even:** the hospital saves from month one whenever its real diesel bill is above **₦117.9 M a month** **ENGINE** (check: fee 69.1 + remaining diesel 48.1 = ₦117.2 M). The real bill is partner data still owed.

**Payback, bought outright only: 1.6 years** incl. the move **ENGINE** — capex ₦3.85 bn + year-5 move ₦678 M (88 % reused) repaid by diesel savings of ≈₦2,840 M a year (check 4,528 ÷ 2,840 = 1.59). Applies to the hospital, not to the investor.

Checks: fuel only, 1,042,790 L × ₦3,277 ÷ 12 = ₦284.8 M a month — the engine's ₦299.2 M adds the generator's operating cost per kWh. Basis of all multi-year figures: ALL energy costs of the protected load — today = grid for 6 h + modelled diesel; service = fee + remaining diesel + remaining grid (so 10 years of service is more than 120 × the fee); no escalation.

## B · ESCO case — the project owner's cash flow

Fee income, capex, O&M, relocation clause, concessional debt. Illustrative structure.

| INPUT | VALUE |
|---|---|
| Energy-as-a-service fee (includes the year-5 move) **ENGINE** | **₦69.1 M/month** |
| Fee income a year **FORMULA** | ₦829.2 M |
| Capex of the system, year 0 **ENGINE** | **₦3,850 M** |
| Relocation in year 5; 88 % of the investment reused **ENGINE** | **₦678 M** |
| Contract term (the engine's 10-year horizon; to be confirmed) | *10 years · editable* |
| O&M and replacement a year | *engine assumptions: concessional debt; editable* |
| Debt service (concessional debt: rate, tenor) | *engine assumptions: concessional debt; editable* |
| ESCO margin | *engine assumption; editable* |

### CUMULATIVE CASH, ₦ M — BEFORE O&M, DEBT SERVICE AND MARGIN **FORMULA**

| YEAR | 0 | 1 | 4 | 5 | 6 | 10 |
|---|---|---|---|---|---|---|
| Fee − capex − relocation | −3,850.0 | −3,020.8 | −533.2 | −382.0 | 447.2 | 3,764.0 |

Fee income over 10 years ₦8,292 M vs capex + relocation ₦4,528 M: **5.5 years of fee** cover both, before O&M, debt service and margin — arithmetic on the inputs, illustrative. The engine's own fee already prices in an annuity on capex + move at a concessional rate over the term, O&M and replacement, and an ESCO margin; those constants are not repeated here.

### FINANCING SOURCES NAMED IN THE PLAN

- The Rural Electrification Agency's public-sector solarisation programme (1.5 MW solar hybrid at the National Orthopaedic Hospital, Dala, Kano)
- The EU and Enugu State ₦4.3 bn Solar for Health programme (25 facilities)
- Development-bank financing for health-facility electrification · ESCOs with concessional debt

Ownership options compared by the engine: hospital buys (upfront capital; a federal asset tied to today's buildings) · equipment lease · energy-as-a-service (recommended: ₦0 upfront; the ESCO owns, maintains and moves the system).

## C · GridRescue AI software case — proposed, illustrative, not a forecast

Software revenue minus operating costs. The investment funds the software business, not hospital hardware.

**Proposed prices** **PROPOSED**: paid site assessment **US$ 2,000 per site** · annual software licence **US$ 3,000 per site per year** · enterprise multi-site and utility licences, integration / white-label fees: not priced · performance-based fee: later, where permitted, after independently verified savings. Recommended hybrid: assessment + licence + enterprise partnerships.

| YEAR-ONE SCENARIO (ILLUSTRATIVE ONLY — NOT FORECASTS, NO CONTRACTS) | 10 SITES | 20 SITES |
|---|---|---|
| Assessment revenue **FORMULA** | US$ 20,000 | US$ 40,000 |
| Licence revenue, first year **FORMULA** | US$ 30,000 | US$ 60,000 |
| **Gross software revenue** | **US$ 50,000** | **US$ 100,000** |
| Operating costs | *to be validated* | *to be validated* |
| Customer acquisition cost | *to be validated* | *to be validated* |
| Renewals in year two | *to be validated* | *to be validated* |
| Operating result | not computed | not computed |
| Investor return | not computed | not computed |

**Investor ask** **PROPOSED**: **US$ 150,000** in two milestone-linked tranches of US$ 75,000 — the second after the simulation engine passes agreed validation tests against hospital data and a pilot partner is secured. Lean pilot option: US$ 75,000, about 6 months. Use of funds on page 2.

**Targets** (planning targets, not achievements): commercial software release in 9–12 months · 2 paid pilot sites by month 9 · 10 paying sites by month 12.

> **Investor return is not computed: no exit, valuation or discount-rate assumptions are made. Simple payback applies to the hospital case when it buys outright, not to the investor.**

Labels: **ENGINE** engine result, modelled · **FORMULA** arithmetic on the inputs · **PUBLISHED** source named · **PROPOSED** to be validated. Workbook: GridRescue_AI_Financial_Model.xlsx — sheets A Hospital case · B ESCO case · C Software case · D Software scenario Y1–Y4 · Use of funds · Assumptions and sources.

# Use of funds · tranches · roadmap · positioning · sources

The proposed seed investment funds the **software business** — not hospital hardware, which needs separate project financing (e.g. an ESCO with concessional debt).

## Use of funds — US$ 150,000

Planning estimates

| LINE | US$ | SHARE |
|---|---|---|
| Software engineering and productionisation | 70,000 | 46.7 % |
| Energy engineering and model validation | 20,000 | 13.3 % |
| Security, testing and hardening | 12,000 | 8.0 % |
| Cloud hosting, weather and geospatial data | 8,000 | 5.3 % |
| Hospital pilot integration and field validation | 15,000 | 10.0 % |
| Customer acquisition and partner onboarding | 10,000 | 6.7 % |
| Legal, contracts and IP | 5,000 | 3.3 % |
| Contingency | 10,000 | 6.7 % |
| **Total** | **150,000** | **100 %** |

### TWO MILESTONE-LINKED TRANCHES

| | |
|---|---|
| **Tranche 1** — at the start | US$ 75,000 |
| **Tranche 2** — after the simulation engine passes agreed validation tests against hospital data and a pilot partner is secured | US$ 75,000 |
| **Lean pilot option** — about 6 months; scope to be agreed | US$ 75,000 |

### SOFTWARE REVENUE SCENARIO Y1–Y4 ILLUSTRATIVE

| US$ | Y1 | Y2 | Y3 | Y4 |
|---|---|---|---|---|
| Paying sites, end of year | 10 | 30 | 65 | 115 |
| New / average sites | 10 / 5 | 20 / 20 | 35 / 47.5 | 50 / 90 |
| **Revenue (new × 2,000 + average × 3,000)** | **35,000** | **100,000** | **212,500** | **370,000** |
| Cumulative revenue | 35,000 | 135,000 | 347,500 | 717,500 |

Site targets from the team's planning model; operating costs to be validated. Cumulative revenue passes the US$ 150,000 seed during Y3 — before operating costs; not a profit or an investor return. Sheet D of the workbook.

## Roadmap, 6–18 months

Months counted from the investment; no calendar dates by design

| | |
|---|---|
| **Months 0–3** | Validate the engine with the hospital's load audit and generator logs; security hardening |
| **Months 3–9** | Two paid pilots (Enugu first) with an ESCO partner and the local distribution company |
| **Months 9–12** | Commercial release; 10 paying sites |
| **Months 12–18** | Utilities and other critical facilities; further outage scenarios (weather groups, cyber/operational contingencies); new countries by adding a tariff table |

### TARGETS (PLANNING TARGETS, NOT ACHIEVEMENTS)

Commercial software release in 9–12 months · 2 paid pilot sites by month 9 · 10 paying sites by month 12.

### WHO PAYS GRIDRESCUE AI

Hospital facilities and engineering teams first; hospital boards; ESCOs and integrators; utilities. Revenue is software: assessments, licences, enterprise and integration fees — never hardware, never the ESCO fee.

### THREE QUESTIONS A JUDGE MAY ASK

- **Why 150k?** The use-of-funds table and the two tranches.
- **Why pay when REopt is free?** The hospital workflow, relocation planning, locally validated data, financing-ready outputs, and an answer in minutes without an expert.
- **How do hospitals save?** The baseline-vs-proposed cost model (case A), software fees separate from hardware, ESCO financing (case B).

## Positioning — checked against the tools judges know

The simulation itself is not claimed as new

- **REopt** (NREL, free): sizes PV, a backup energy system and a generator so a site survives a specified outage; US-based with guidance for international sites; built for energy experts.
- **HOMER Pro** (UL Solutions, commercial): annual microgrid simulation and least-cost design; priced for consultants, steep learning curve.
- **HOMER Powering Health Tool** (World Bank/ESMAP, free): least-cost solar + backup energy system + diesel combinations for clinics and hospitals — a sizing and cost tool, not an outage-by-outage simulation; no relocation, no grid mapping, no ESG report.
- **Odyssey Energy Solutions**: project planning, financing, procurement and monitoring for developers.
- **MapYourGrid / OpenStreetMap community**: grid infrastructure mapping (our data partner, not a competitor).

### WHAT NONE OF THEM DOES IN ONE FLOW, AND GRIDRESCUE AI DOES TODAY

1. From a map pin to an investment-ready hospital plan in minutes, on open data, with no API keys and no expert.
2. The grid mapper with the "hospital is moving" new-site lens: substations and lines within 50 and 100 km, published supply hours, areas ranked for the new building — then a relocation plan with a move cost and a reuse percentage.
3. Every energy-system family simulated hour by hour on the site's own sun, grid hours and load, four stress tests, with a vendor-neutral ML classification of the energy system that the simulation verifies.
4. Local market assumptions built in: the NBS diesel price, the state tariff, Nigerian funders and programmes.
5. The ESG report (Scope 1 and Scope 2, E·S·G lines) and the before → after KPIs written before anything is bought.

Claim the integrated, hospital-specific, relocatable, financeable workflow — never the simulation itself.

**Sources and labels: NBS Automotive Gas Oil price watch (diesel, national average) · EERC Band A tariff · NASA POWER, PVGIS, PVWatts, Global Solar Atlas (sun) · OpenStreetMap (ODbL, © OpenStreetMap contributors) for the grid read · IPCC default 2.68 kg $CO_2$/L · Problem Brief 12 for the hospital's reported facts · all other figures are engine results of the running application (modelled, ±30 %) or planning estimates. Vendor-neutral: no brand names; GridRescue AI sells no hardware.**